

Tutorial Sheet-8

Indian Institute of Technology Jodhpur

MAL1020 (Mathematics-II)

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1. Find the general solution of the following differential equations:

(i) $(D^2 + 2D + 2)y = 4e^{-x} \sec^3(x)$

(ii) $(x^2 D^2 + xD - 1)y = \frac{1}{x^2}$

(iii) $y'' + 8y' + 16y = 64 \cosh(4x)$

(iv) $(D^2 + 1)y = 12 \cos^2 x$

(v) $(D^2 + 4D + 4)y = \frac{e^{-2x}}{x^2}$

(vi) $(D^2 - 2D + 1)y = 3e^x x^{\frac{3}{2}}$

(vii) $y'' - 4y' + 5y = e^{2x} \operatorname{cosec}(x)$

(viii) $y'' + 2y' - 35y = 12e^{5x} + 37 \sin(5x)$

2. Apply Picard's iteration method to solve the following problems.

(a) $y' = y^2, \quad y(0) = 1$

(b) $y' = y - y^2, \quad y(0) = \frac{1}{2}$

(c) $y' = xy + 2x - x^3, \quad y(0) = 0$

3. Solve the following differential equations:

(i) $(x^2 D^2 + 1)y = 3x^2$

(ii) $(xD^3 + D^2)y = 1/x$.

(iii) $(x^2 D^2 - 3xD + 4)y = 2x^2$.

(iv) $(x^2 D^2 - 2)y = x^2 + (1/x)$

(v) $(x^2 D^2 - 3xD + 5)y = \sin(\log x)$.

(vi) $(3x^2 D^2 - 5xD + 5)y = \sin(\log x)$.

(vii) $x^3 (d^3 y/dx^3) + 3x^2 (d^2 y/dx^2) + x(dy/dx) + 8y = 65 \cos(\log x)$.

(viii) $x^4 (d^4 y/dx^4) + 6x^3 (d^3 y/dx^3) + 4x^2 (d^2 y/dx^2) - 2x(dy/dx) - 4y = 2 \cos(\log x)$.

4. Apply Picard's method to solve the following initial value problem up to the third approximation: $dy/dx = 2y - 2x^2 - 3$ given that $y = 2$ when $x = 0$.

5. Using Picard's method of successive approximations, find the third approximation of the solution of the equation: $dy/dx = x + y^2$, where $y = 0$ when $x = 0$.